

EC1001-1 – BASIC ELECTRONICS

Max. Marks:100

Off-Campus
First Semester
3 Hours

2) Assume missing data suitably.

20 Marks

- 1-

13. In Colpitts' oscillator, the components used in the feedback network are
 A) 2L and 1C B) 2C and 1L
 C) 2R and 2C D) 2L and 2C
14. With a resistance value of $R=1k\Omega$ in a feedback network of RC oscillator, frequency of oscillations generated is 5 kHz. The value of the capacitor C is
 A) $0.129 \mu F$ B) $0.0219 \mu F$
 C) $129 \mu F$ D) $0.0129 \mu F$
15. Gain with negative feedback is given by $A_f = \frac{A}{1+A\beta}$. The closed loop gain is
 A) A_f B) A
 C) β D) None of these
16. An amplifier has an open loop voltage gain of 1000. If 10% negative voltage series feedback is used, then the closed loop gain is
 A) 99.9 B) 9.9
 C) 0.9 D) 990
17. Which of the following statements are true for von Neumann architecture?
 A) Separate bus between the program memory and data memory B) External bus for program memory and data memory
 C) External bus for data memory only D) Shared bus between the program memory and data memory
18. Harvard architecture has _____
 A) Dedicated buses for data and program memory B) Pipeline technique
 C) Complex architecture D) All of these
19. The unit used for measuring message or information is
 A) Hertz B) Ohms
 C) Bits per second D) Meter per second
20. The inherent interference resistance property between wireless cellular channels is observed in
 A) Frequency Division Multiple Access B) Time Division Multiple Access
 C) Code Division Multiple Access D) Space Division Multiple Access

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat circuit diagram, explain the principle of operation of a full wave bridge rectifier. Draw the relevant waveforms.	6	L2	1	1
	b) Define efficiency and ripple factor of a rectifier. Deduce the maximum values of both parameters for a half wave rectifier.	6	L2	1	1
	c) A 5V regulated power supply is required to produce from a 12V direct current (DC) power supply input source. The maximum power rating P_z of the Zener diode is 2W. Calculate: i) - The maximum current following through the Zener diode, ii) The minimum value of the series resistor, R_s , iii) The load current I_L , if a load resistor of $1k\Omega$ is connected across the Zener diode, iv) The Zener current I_z at full load.	4	L3	1	1
2.	a) Explain D.C. load line analysis of a Bipolar Junction Transistor (BJT) in Common Emitter (CE) configuration.	6	L2	2	1
	b) Calculate α and β for a transistor with collector current of 1 mA, base current of $25 \mu A$. Determine the new value of base current to give a collector current of 5 mA.	6	L3	2	1
	c) With a neat circuit diagram, explain how BJT can be made to operate as a switch.	4	L2	2	1

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| 3. | a) | With the neat construction diagrams, explain the drain characteristics of n-channel JFET. | 6 | L2 | 2 | 1 |
| | b) | For a Junction Field Effect Transistor (JFET), the $I_{DSS} = 6\text{mA}$, $V_p = -4.5\text{V}$. Determine the drain current I_D for the following cases:
i) $V_{GS} = -2\text{V}$
ii) $V_{GS} = -4\text{V}$. | 6 | L3 | 2 | 1 |
| | c) | Explain the working of a CMOS inverter with a neat circuit diagram. | 4 | L2 | 2 | 1 |

Unit – II

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| 4. | a) | With the help of a neat circuit diagram, derive the expression for output voltage of an inverting amplifier circuit using Op-amp. Draw the input and output waveforms considering the input voltage of 2V peak and a gain of 4. Assume the V^+ and V^- as $\pm 12\text{V}$. | 6 | L2 | 3 | 1 |
| | b) | Design a summing amplifier using Op-amp, to get an output voltage of $V_o = -(3V_1 + 4V_2 + 5V_3)$, assuming a feedback resistor of $120\text{k}\Omega$. Draw the circuit. | 6 | L3 | 3 | 1 |
| | c) | Draw the equivalent circuit of an Op-Amp and write the significance of each parameter in it. | 4 | L1 | 3 | 1 |
| 5. | a) | With the help of a neat circuit diagram, derive the expression for the output voltage of an integrator circuit using Op-amp. | 6 | L2 | 3 | 1 |
| | b) | Draw the circuit of a IC 555 timer as an oscillator in astable mode. For an IC 555 based oscillator in astable mode, operating at a duty cycle of 75%, with frequency of operation 1kHz , the values of $R_2 = 3.6\text{k}\Omega$ and $C = 0.1\mu\text{F}$. Calculate the value of the resistor R_1 . | 6 | L3 | 3 | 1 |
| | c) | An amplifier has an open loop voltage gain of 1000. If the feedback factor is 10%, find the closed loop voltage gain. | 4 | L3 | 4 | 1 |
| 6. | a) | State Barkhausen's criterion for generating sustained oscillations. Derive the conditions with the help of an oscillator block diagram. | 6 | L2 | 4 | 1 |
| | b) | With a neat circuit diagram, explain the operation of a RC phase shift oscillator. Analyse the role of RC components as feedback network. | 6 | L2 | 4 | 1 |
| | c) | In a Colpitts oscillator, $C_1 = 100\text{pF}$; $C_2 = 260\text{pF}$. Find the value of L if the frequency of oscillation is 40kHz . | 4 | L3 | 4 | 1 |

Unit – III

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| 7. | a) | What is meant by modulation in communication system? Write the needs for modulation. | 6 | L2 | 5 | 1 |
| | b) | Explain the following concepts of a cellular system:
i) Frequency reuse
ii) Capacity of a cluster. | 6 | L2 | 5 | 1 |
| | c) | Explain different control and voice channels available between a mobile unit and base station for the initiation of a call in a cellular system. | 4 | L2 | 5 | 1 |
| 8. | a) | With a neat diagram, explain the various elements of an Embedded system. | 6 | L2 | 5 | 1 |
| | b) | Differentiate between Microprocessor and Microcontroller. | 6 | L2 | 5 | 1 |
| | c) | Discuss the optocoupler with a diagram. Diagrammatically show the usage of optocoupler. | 4 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
